NUMERICAL COMPUTING II

Integrated Numerical Methods Project

Application of Numerical Methods to a Temperature Time-Series Dataset

1. Aim

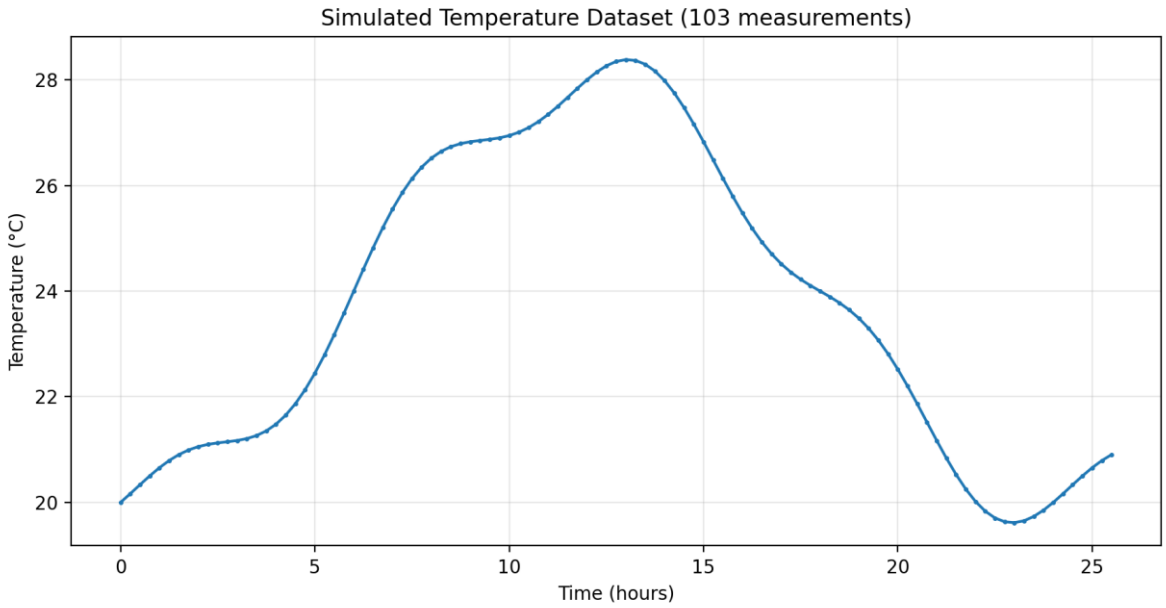
The aim is to apply numerical methods studied in Numerical Computing II to one dataset and compare their results. The methods included are Lagrange interpolation, forward/backward/central finite differences, Richardson extrapolation, the trapezoidal rule, Simpson's 1/3 rule, and Simpson's 3/8 rule.

2. Dataset and data source

For a reproducible demonstration, this report uses a simulated temperature time series with 103 measurements from 0 to 25.5 hours, sampled every 0.25 hour (15 minutes). The profile is $T(t) = 24 + 4\sin(2\pi(t-6)/24) + 0.6\sin(2\pi t/6)$, where T is temperature in °C and t is time in hours. This is a synthetic teaching dataset, not a record of actual weather observations. The formula is stated so the results can be reproduced and an exact derivative and integral can be used to assess numerical error.

A real alternative is the National Water Data Portal's hourly temperature telemetry dataset from CWPRS, which describes hourly temperature records from meteorological stations in Maharashtra. If a real-data-only submission is required, replace the simulated values with a downloaded CSV and rerun the same methods.

Dataset file: temperature_dataset_103_points.csv (provided alongside this report).



3. Methods and formulas

3.1 Lagrange interpolation

$P(x) = \Sigma\, f(x_i)L_i(x)$, where $L_i(x) = \Pi_{j\neq i}\,(x-x_j)/(x_i-x_j)$.

A four-point local Lagrange polynomial was formed using the four data points nearest to t = 7.3 hours. It estimated T(7.3) = 25.92207571 °C; the value from the known generating function is 25.92211600 °C, giving absolute error 4.029e-05 °C.

3.2 Numerical differentiation

Forward difference: $D_+(h) = [T(t+h) - T(t)]/h$.

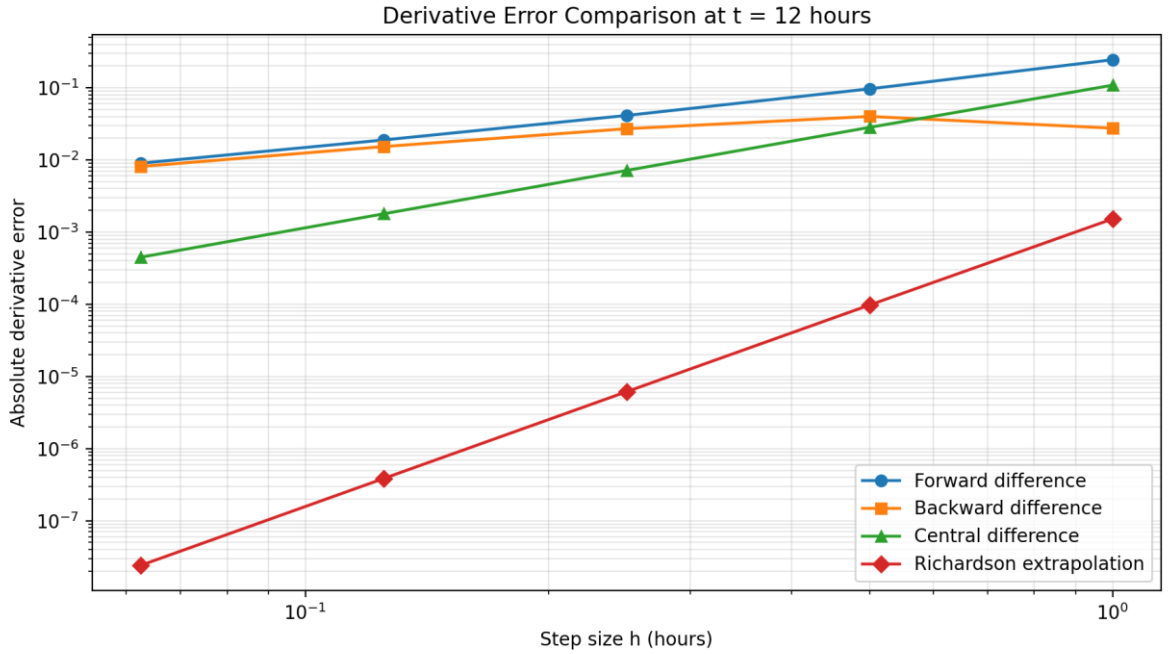
Backward difference: $D_-(h) = [T(t) - T(t-h)]/h$.

Central difference: $D_0(h) = [T(t+h) - T(t-h)]/(2h)$.

Richardson extrapolation: $R(h) = [4D_0(h/2) - D_0(h)]/3$.

These methods were compared at t = 12 hours. The exact derivative from the generating function is 0.62831853 °C/hour.

h	Forward	Error	Backward	Error	Central	Error	Richardson	Error
1	0.383319	2.45e-01	0.655912	2.76e-02	0.519615	1.09e-01	0.626795	1.52e-03
0.5	0.531559	9.68e-02	0.668441	4.01e-02	0.600000	2.83e-02	0.628221	9.76e-05
0.25	0.586908	4.14e-02	0.655423	2.71e-02	0.621166	7.15e-03	0.628312	6.14e-06
0.125	0.609393	1.89e-02	0.643659	1.53e-02	0.626526	1.79e-03	0.628318	3.84e-07
0.0625	0.619303	9.02e-03	0.636437	8.12e-03	0.627870	4.48e-04	0.628319	2.40e-08



4. Summary of findings

The local Lagrange estimate at 7.3 hours differed from the known profile by 4.029e-05 °C. Differentiation results show the effect of step size: forward and backward differences generally have first-order truncation error, central difference has second-order truncation error, and Richardson extrapolation cancels the leading second-order term to give fourth-order truncation error for a sufficiently smooth function. At very small step sizes, round-off can become important.

For integration, the trapezoidal estimate was 606.72479677 °C·hour, Simpson's 1/3 estimate was 606.72600012 °C·hour, and Simpson's 3/8 estimate was 606.72601850 °C·hour. Their absolute errors against the analytic integral were 1.189e-03, 1.448e-05, and 3.286e-05 °C·hour, respectively.

5. Conclusions

A single temperature time-series dataset can be used to demonstrate several numerical methods. Lagrange interpolation estimates values between data points; finite differences estimate the rate of change; Richardson extrapolation can improve derivative accuracy; and the trapezoidal and Simpson rules estimate the integral over time. The most suitable method depends on the spacing of the data, smoothness of the function, required accuracy, and the computational effects of round-off error.

Limitation: this report uses simulated rather than observed temperatures so that the calculations are transparent and can be checked against a known function. The National Water Data Portal provides real hourly temperature datasets if the instructor requires observed data.

6. Dataset preview

Time (hours)	Temperature (°C)
0.00	20.00000
0.25	20.16386
0.50	20.33422
0.75	20.50112
1.00	20.65591
1.25	20.79183
1.50	20.90448
1.75	20.99206
2.00	21.05551
2.25	21.09839

The accompanying CSV contains all 103 measurements. Source for a real-data alternative: National Water Data Portal,

Link for the datasets:
“Temperature (Telemetry Hourly), CWPRS” — https://nwdp.nwic.gov.in/en/dataset/temperature-telemetry-hourly-cwprs

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